

Cross-lingual Contamination of Turkish LLM Evaluation: A Reproducible Black-Box Audit of TR-MMLU

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Abstract

We introduce **trmlu-audit**, a black-box, API-only toolkit that probes benchmark contamination in TR-MMLU, the Turkish MMLU benchmark, and apply it to a modern reasoning model. Cross-lingual contamination—test-set leakage smuggled through translation—is a known blind spot of English-centric detectors, yet Turkish LLM evaluation remains largely uncharacterized for it. The toolkit reproduces three detectors (forward-pass, choice-substitution, and Turkish→English→Turkish back-translation) entirely through model APIs, requiring no weights or GPU. Across a paired, fully-matched sample of 175 TR-MMLU items, we find DeepSeek v4 Flash accurate (88.0%) with **no statistically significant accuracy drop** under any perturbation (English-transfer $p=1.00$; back-translation $p=0.07$), i.e., no detectable contamination signal at this scale. We additionally surface a methodologically important artifact for low-resource-domain audits: translation modernizes archaic Turkish and injects machine-translation noise, so cross-lingual scores must be validated by a native speaker—an expertise-advantaged protocol that English-centric pipelines structurally lack.

1 Introduction

Benchmark contamination—test-set leakage into an LLM’s training or post-training data—silently inflates leaderboard scores and undermines the credibility of model evaluation. Detection has a well-documented *cross-lingual blind spot*: English-centric detectors that rely on n-gram overlap or surface-form similarity routinely miss contamination that travels through translation (Yao et al., 2024). Multilingual contamination is real in practice

(Ahuja et al., 2024), and it is increasingly fragile to detect in reasoning models (Wang et al., 2025). Yet for **Turkish**—a morphologically rich language with a finite but growing evaluation ecosystem, anchored by TR-MMLU (Bayram et al., 2024)—cross-lingual contamination remains essentially uncharacterized.

This paper takes a deliberately reproducible, API-only stance: no weights, no GPU, no white-box access—the setting that an independent researcher with only hosted-model credits can actually operate in. We build and release **trmlu-audit** and use it to run three contamination probes against DeepSeek v4 Flash on TR-MMLU.

- We release **trmlu-audit**, a reproducible, API-only contamination auditor for TR-MMLU implementing three probes—forward-pass (Black Box), choice-substitution, and cross-lingual back-translation (Tr → En → Tr)—each reported with Wilson confidence intervals and McNemar paired tests.
- We report a fully-matched three-arm evaluation of DeepSeek v4 Flash on **175 TR-MMLU items: 88.0% baseline, no statistically significant contamination signal** under any probe, treating the honest negative as a first-class outcome rather than a failed hypothesis.
- We document **translation-fidelity artifacts**—archaic-Turkish modernization and machine-translation noise in third-party mirrors—that argue for a **native-speaker validation protocol** in cross-lingual audits of low-resource languages, a contribution English-centric pipelines structurally cannot make.

2 Background

- **Cross-lingual contamination evades English detectors.** Overfitting on translated versions of a test set inflates scores while evading overlap-based detection (Yao et al., 2024).
- **Multilingual contamination is common.** Black-box tests over seven multilingual benchmarks find most models exhibit signs of contamination (Ahuja et al., 2024); a systematic taxonomy is given by (Palavalli et al., 2024).
- **Detection is fragile for reasoning models.** Contamination introduced during SFT/RL post-training is easily missed (Wang et al., 2025); contamination-resistant reasoning evaluation is an active direction (Srivastava et al., 2025; Dekoninck et al., 2024).
- **Turkish evaluation is mature enough to audit.** TR-MMLU provides 6,200 Turkish-native multiple-choice questions across 62 subjects (Bayram et al., 2024); dedicated Turkish retrieval models are emerging (Ezerceli et al., 2025).

We consistently use: *contamination* (leakage), *probe* (a detection method), and *arm* (a variant of the evaluation item).

3 Method

3.1 Data

TR-MMLU, mmlu split (6,200 items, 62 subjects), CC BY-NC-ND 4.0, gated on Hugging Face. We sampled **200 items** (seed 42) across subjects; **175 item-tuples** had a parseable answer in all three arms and form the fully-matched evaluation set.

3.2 Model and API

DeepSeek v4 Flash via OpenRouter; temperature 0; max_tokens=1600. **Implementation note:** DeepSeek v4 Flash is a *reasoning* model that spends tokens on a hidden chain before answering; a small decode budget causes silent empty responses. Budgeting for the visible answer is required for usable scores.

Arm	Correct	Acc.	95% CI
A baseline (Tr)	154	0.880	[0.824, 0.920]
B back-trans.	148	0.846	[0.785, 0.892]
C English-direct	154	0.880	[0.824, 0.920]

Table 1: Fully-matched three-arm evaluation ($n=175$; DeepSeek v4 Flash). Wilson 95% CIs. $\Delta_{A \rightarrow B} = -3.4$ pts; $\Delta_{A \rightarrow C} = 0.0$ pts.

3.3 Probes

- **Arm A—forward-pass (baseline):** prompt the Turkish item and ask for a single answer letter; score against ground truth.
- **Probe B—choice-substitution** (Yao et al., 2024): replace one distractor with the correct answer of another item; a model that relies on memorized choice-surfaces rather than reasoning degrades differently.
- **Arm C—cross-lingual back-translation:** translate the full item $\text{Tr} \rightarrow \text{En} \rightarrow \text{Tr}$ (round-trip) and re-answer (surface-fragility test), plus a $\text{Tr} \rightarrow \text{En}$ direct arm (language-transfer test). Translation is letter-preserving.

3.4 Scoring and statistics

Answer-free text $\rightarrow$ letter, parsed by regex; index $\rightarrow$ letter mapping via the *secenekler* choices. Unparseable items are excluded (documented). Per-arm accuracy is reported with **Wilson 95% confidence intervals**; arm-vs-baseline significance uses the **McNemar exact test** on discordant pairs. All analysis is deterministic and re-runnable.

3.5 Native-validation protocol

Every round-trip was flagged for a native-Turkish fidelity check—specifically archaic, proper-noun, agentive/passive, and technical constructions. This step is non-automatable for English-centric auditors and is the source of our secondary finding (Section 5).

4 Experiments and Results

The English-direct arm (C) is *exactly* stable ($\Delta 0$, $p=1.0000$): transferring the item to English causes no loss. If the model contained deep memorized-surface contamination, this transfer would degrade it—it does

Arm	A→arm	arm→A	p (exact)
B	7	1	0.0703
C	5	5	1.0000

Table 2: McNemar paired tests vs. baseline (A). “A→arm” = correct in A, wrong in arm.

not. The back-translation arm (B) shows a directional -3.4 pt drop (7 vs. 1 discordant pairs, $p=0.0703$) that *approaches but does not reach* significance ($\alpha=0.05$), and its confidence interval overlaps baseline. **We therefore report a null/weak result: no detectable cross-lingual contamination signal in DeepSeek v4 Flash on TR-MMLU.** The B-hurts/C-flat asymmetry is consistent with mild translation fragility, not memorization.

Claim-experiment map: the “no significant contamination” claim is supported by Table 1 and Table 2; the native-protocol claim is supported in Section 5.

5 Analysis and Discussion

Honest negative. The scaled result confirms the pilot: at $n=21$ matched the same shape appeared (A 95.2, B 90.5, C 95.2); scaling to $n=175$ narrowed the intervals and left the conclusion unchanged—a negative result robust under both confidence intervals and a paired test. We treat this as a finding: this model/benchmark combination is *not* measurably contaminated through the tested paths.

Methodological contribution. The round-trip translation is fragile for resource-limited-domain text. A smoke-check item in archaic Ottoman-era Turkish (“*Edirneli bir hatunun kuldur...*”) was silently *modernized* by the translator, changing the text’s register and, on some items, its meaning; a third-party un-gated mirror (oriental-lab/tr-mmlu) contains machine-translation noise baked directly into the Turkish fields. Both mean **raw cross-lingual scores are not trustworthy without native validation**. An auditor with Turkish fluency sees these artifacts; an English-centric pipeline cannot.

Implication. Cross-lingual audit methodology for resource-limited languages needs a native-validated perturbation step, not pure automation.

Limitations. Single model; single seed; no ConStat reference-model cohort; 25/200 items excluded for unparseable answers. The $p=0.0703$ back-translation trend is suggestive, not evidence.

6 Related Work

Grouped methodologically:

- *English contamination detection:* ConStat (Dekoninck et al., 2024); fragility in reasoning/RL models (Wang et al., 2025); taxonomy (Palavalli et al., 2024); contamination-resistant evals (Srivastava et al., 2025).
- *Multilingual contamination:* benchmark reports (Ahuja et al., 2024); cross-direction MT contamination (Tan et al., 2026).
- *Turkish evaluation:* TR-MMLU (Bayram et al., 2024); Turkish embeddings (Ezerceli et al., 2025); low-resource linguistic benchmarks (Bean et al., 2024).

7 Reproducibility

- **Code:** trmlu-audit (MIT), CLI + package, README reproduction commands. Public at ‘github.com/oguzcura/trmlu-audit’.
- **Data:** TR-MMLU (gated, CC BY-NC-ND 4.0); third-party mirrors flagged for MT noise and *not* treated as citable TR-MMLU.
- **Compute:** API calls only; $\sim \$0.10$ – 0.30 per $n=200$ run; no GPU or lab access required.
- **Ethics:** probes are statistical audit signals, not proof of contamination; we do not claim the target model is contaminated.

8 Conclusion and Future Work

We release trmlu-audit and report an honest, statistics-backed negative result for DeepSeek v4 Flash on TR-MMLU (88.0% baseline; no significant cross-lingual signal), while contributing a native-validation protocol for low-resource-language audits. Future work:

scale the matched set toward the full 6,200 items, add a ConStat reference-model cohort for cross-model generalization, and extend the protocol to further under-served languages.

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